Functions: Modulus & Inverse

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Modulus Equations & Linear Graphs**

The complex number z is defined by $z = \frac{a+3\mathrm{i}}{2+a\mathrm{i}}$, where a is a real constant. Given that $\arg(z) = \frac{\pi}{4}$, find the value of a.

To find a, we first express z in Cartesian form $x + \mathrm{i}y$ by multiplying the numerator and denominator by the complex conjugate of the denominator, $2 - a\mathrm{i}$: $z = \frac{(a+3\mathrm{i})(2-a\mathrm{i})}{(2+a\mathrm{i})(2-a\mathrm{i})} = \frac{2a - a^2\mathrm{i} + 6\mathrm{i} - 3a\mathrm{i}^2}{4+a^2} = \frac{5a + (6-a^2)\mathrm{i}}{4+a^2}$ Thus, the real and imaginary parts are: $\mathrm{Re}(z) = \frac{5a}{4+a^2}$, $\mathrm{Im}(z) = \frac{6-a^2}{4+a^2}$ For.

2 **[Foundational]** • **Target: Modulus Inequalities**

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

1. Find the parameter k: By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k$ $6 + k = 4 \Rightarrow k = -2$ 2. Determine the inverse function $f^{-1}(x)$: With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y+2}{3}$ Thus, $f^{-1}(x) = \frac{x+2}{3}$. 3. Solve the equation $f^{-1}(x) = g(x)$: $\frac{x+2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$ Since the ri.

3 **[Foundational]** • **Target: Composite and Inverse Functions in A2**

The random variable X has a Poisson distribution with parameter λ, where $\lambda > 0$. Given that $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, find the value of λ.

For a random variable $X \sim \mathrm{Po}(\lambda)$, the probability mass function is given by: $\mathrm{P}(X = x) = \frac{\mathrm{e}^{-\lambda}\lambda^x}{x!}$, $x = 0, 1, 2, \ldots$ Given $\mathrm{P}(X = 3) = \mathrm{P}(X = 4)$, we substitute $x = 3$ and $x = 4$: $\frac{\mathrm{e}^{-\lambda}\lambda^3}{3!} = \frac{\mathrm{e}^{-\lambda}\lambda^4}{4!}$ Since $\mathrm{e}^{-\lambda} > 0$ and $\lambda > 0$, we can divide both sides by $\mathrm{e}^{-\lambda}\lambda^3$: $\frac{1}{3!} = \frac{\lambda}{4!}$ Multiplyi.

4 **[Foundational]** • **Target: Transformations & Analytical Modeling**

Given that $\cos\theta = -\frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\sin 2\theta$.

We are given $\cos\theta = -\frac{3}{5}$ for an angle in the second quadrant, $\frac{\pi}{2} < \theta < \pi$. 1. In the second quadrant, the sine function is positive: $\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \left(-\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. 2. Use the double-angle identity for sine, $\sin 2\theta = 2\sin\theta\cos\theta$: $\sin 2\theta = 2\left(\frac{4}{5}\right)\left(-\frac{3}{5}\right) = -\frac{24}{25}$. Common misconceptions.

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5 [Foundational] • Target: Modulus Equations & Linear Graphs

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

To solve the modulus equation $|3x - 2| = 2|x + 1|$, we can either consider the linear cases or square both sides. Method 1: Considering linear cases The equation $|A| = |B|$ (where $B = 2(x + 1)$) is equivalent to $A = B$ or $A = -B$: - Case 1: $3x - 2 = 2(x + 1) \Rightarrow 3x - 2 = 2x + 2 \Rightarrow x = 4$ - Case 2: $3x - 2 = -2(x + 1) \Rightarrow 3x - 2 = -2x - 2 \Rightarrow 5x = 0 \Rightarrow x = 0$ Method 2: Squaring both sides Since both sides are non-negative, we can square both sides.

6 [Foundational] • Target: Modulus Inequalities

A machine produces steel rods whose lengths are normally distributed with standard deviation $\sigma = 2.4$ cm. A quality controller suspects that the mean length has increased from the target value of 50 cm. A random sample of 36 rods is measured, yielding a sample mean of $\bar{x} = 50.8$ cm. Find the value of the standardized test statistic z for testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.

Under the null hypothesis $H_0 : \mu = 50$, the sample mean of $n = 36$ independent observations follows a normal distribution: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(50, \frac{2.4^2}{36}\right)$ The standard error of the mean is: $\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{2.4}{\sqrt{36}} = \frac{2.4}{6} = 0.4$ The standardized test statistic z is calculated as: $z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{50.8 - 50}{0.4} = \frac{0.8}{0.4} = 2.0$ Common errors include: - O.

7 [Foundational] • Target: Composite and Inverse Functions in A2

The function f is defined for all real numbers by $f(x) = x^3 + 3x - 5$. Between which pair of consecutive integers does the unique real root of the equation $f(x) = 0$ lie?

To locate the root using the change of sign method, evaluate $f(x)$ at consecutive integer values: $f(0) = (0)^3 + 3(0) - 5 = -5 < 0$ $f(1) = (1)^3 + 3(1) - 5 = 1 + 3 - 5 = -1 < 0$ $f(2) = (2)^3 + 3(2) - 5 = 8 + 6 - 5 = 9 > 0$ Since $f(x)$ is a continuous polynomial function and there is a sign change between $x = 1$ (where $f(1) < 0$) and $x = 2$ (where $f(2) > 0$), by the Intermediate Value Theorem, a root of $f(x) = 0$ lies between 1 and 2. Checking the other intervals: - On $[0, 1]$, $f(0) = -5$ and.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Transformations & Analytical Modeling

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $P(1 \le X \le 2)$.

To find the probability $P(1 \le X \le 2)$, we integrate the probability density function $f(x)$ over the interval $[1, 2]$: $P(1 \le X \le 2) = \int_1^2 \frac{3}{8}x^2 \, dx$ Find the antiderivative: $\int \frac{3}{8}x^2 \, dx = \frac{3}{8} \cdot \frac{x^3}{3} = \frac{x^3}{8}$ Evaluate between the limits $x = 1$ and $x = 2$: $\left[\frac{x^3}{8}\right]_1^2 = \frac{2^3}{8} - \frac{1^3}{8} = \frac{8}{8} - \frac{1}{8} = \frac{7}{8}$ Common errors include calculating t.

9 [Intermediate] • Target: Modulus Equations & Linear Graphs

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

To find the gradient of the tangent to the curve $y = \ln(2x^3 + 1)$ at $x = 1$, we differentiate y with respect to x using the chain rule. Let $u = 2x^3 + 1$, so $y = \ln(u)$. Then: $\frac{du}{dx} = 6x^2 \quad \text{and} \quad \frac{dy}{du} = \frac{1}{u}$ Applying the chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{6x^2}{2x^3 + 1}$ Now evaluate the derivative at $x = 1$.

10 [Intermediate] • Target: Modulus Inequalities

The first term of a geometric progression is 16 and the fourth term is 2. Find the sum to infinity, S_∞, of the progression.

The nth term of a geometric progression is given by $u_n = ar^{n-1}$, where a is the first term and r is the common ratio. Given $a = 16$ and the fourth term $u_4 = 2$: $ar^3 = 2 \Rightarrow 16r^3 = 2 \Rightarrow r^3 = \frac{2}{16} = \frac{1}{8} \Rightarrow r = \frac{1}{2}$ Since $|r| < 1$, the sum to infinity exists and is given by: $S_\infty = \frac{a}{1-r} = \frac{16}{1-\frac{1}{2}} = \frac{16}{\frac{1}{2}} = 32$ Common errors include setting $u_4 = ar^4$ which yields $r^4 = \frac{1}{8}$, or miscalcula.

11 [Intermediate] • Target: Composite and Inverse Functions in A2

Evaluate $\int_0^3 2x(x^2 + 9)^4 \, dx$.

Let $u = x^2 + 9$, $du = 2x \, dx$. $\int u^4 du = \frac{u^5}{5}$. At $x = 3$: $u = 18$, giving 1889568. At $x = 0$: $u = 9$, giving 59049. Result: $\frac{1889568 - 59049}{5} = \frac{1830519}{5}$.

12 [Intermediate] • Target: Transformations & Analytical Modeling

Find the set of all real solutions to the equation $\log_2(x - 1) - \log_4(x + 14) = 1$

To solve the equation, we first express all logarithms in the same base using the change of base formula: $\log_4(x + 14) = \frac{\log_2(x+14)}{\log_2 4} = \frac{\log_2(x+14)}{2} = \frac{1}{2}\log_2(x + 14)$ Substitute this back into the original equation: $\log_2(x - 1) - \frac{1}{2}\log_2(x + 14) = 1$ Multiply through by 2: $2\log_2(x - 1) - \log_2(x + 14) = 2$ Apply the power law and the quotient law of logarithms.

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13 [Intermediate] • Target: Modulus Equations & Linear Graphs

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$
Step 2: Find the parameter k using the given total area of 8.
$\text{Area} = \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8:
$4\sqrt{3k+4} - 8 = 8 \Rightarrow 4\sqrt{3k+4} = 16 \Rightarrow \sqrt{3k+4} = 4$ Squaring both.

14 [Intermediate] • Target: Modulus Inequalities

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

We solve the problem in three steps: Step 1: Determine q using perpendicularity The direction vectors of l_1 and l_2 are $\mathbf{d}_1 = (1; -2; 2)$ and $\mathbf{d}_2 = (2; 3; q)$. Since $l_1 \perp l_2$, their scalar (dot) product must equal zero: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(2) + (-2)(3) + (2)(q) = 0\ 2 - 6 + 2q = 0 \Rightarrow 2q = 4 \Rightarrow q = 2.$ Step 2: Find the parameters λ and μ at the intersection Equating the coordinates of l_1 and l_2: - x.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 10.01 Numerical Methods

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour. Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

1. Find the parameter λ for a 1-hour period: Let X be the number of calls received in 1 hour. Then $X \sim \text{Po}(\lambda)$. The probability mass function is $\text{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$. We are given: $\text{P}(X = 1) = 1.5\,\text{P}(X = 3)$ Substitute the formula for each probability.

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16 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 11.01 Differential Equations

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length 1 km and the second section has length 2 km. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X + Y = 3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X + Y = 3)$.

Let $X \sim \mathrm{Po}(\lambda)$ and $Y \sim \mathrm{Po}(2\lambda)$ be independent Poisson random variables. The sum $T = X + Y$ follows a Poisson distribution with parameter $\lambda + 2\lambda = 3\lambda$. The conditional distribution of X given $X + Y = n$ is binomial $\mathrm{B}\left(n, \frac{\lambda}{\lambda + 2\lambda}\right) = \mathrm{B}\left(n, \frac{1}{3}\right)$. Here, $n = 3$, so $X \mid (X + Y = 3) \sim \mathrm{B}\left(3, \frac{1}{3}\right)$. We want to find.

17 [Challenging] • Target: Modulus Equations & Linear Graphs • Cross-Lesson: 12.01 Poisson Distribution

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

A) The standard error of the mean (SE) is calculated using the formula: $SE = \frac{s}{\sqrt{n}}$, where s is the sample standard deviation and n is the sample size. Given $s = 3.5$ grams and $n = 50$. $SE = \frac{3.5}{\sqrt{50}} \approx \frac{3.5}{7.071} \approx 0.495$ grams. B) To construct a 95% confidence interval for the mean weight, we use the formula: $CI = \bar{x} \pm z^* \times SE$, where $\bar{x}$ is the sample mean and z^* is the critical z-value for a 95% confidence level. For a 95% confidence lev.

18 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 13.01 Continuous Random Variables

The curve C has equation $y = \frac{ax + 6}{x^2 + 3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

To find the unknown parameter a, differentiate y with respect to x using the quotient rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{a(x^2 + 3) - (ax + 6)(2x)}{(x^2 + 3)^2} = \frac{3a - 12x - ax^2}{(x^2 + 3)^2}$ At a stationary point, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, which requires the numerator to be zero: $3a - 12x - ax^2 = 0$ Given that $x = 1$ is a stationary point: $3a - 12(1) - a(1)^2 = 0 \Rightarrow 2a - 12 = 0 \Rightarrow a = 6$ Substitute $a = 6$ back into the numerator of the derivative.

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19 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 14.01 Sampling & Estimation

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

We solve the problem in two steps: Step 1: Find the value of k The Newton-Raphson iteration formula is: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ Differentiating $f(x) = x^3 - 2x^2 - kx + 12$ with respect to x: $f'(x) = 3x^2 - 4x - k$ Evaluating $f(x)$ and $f'(x)$ at $x_0 = 2$: $f(2) = 2^3 - 2(2^2) - 2k + 12 = 8 - 8 - 2k + 12 = 12 - 2k$ $f'(2) = 3(2^2) - 4(2) - k = 12 - 8 - k = 4 - k$ Using the iteration formula for x_1: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow 4 = 2 - \frac{12-2k}{4-k}$ Rearranging.

20 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 15.01 Hypothesis Testing

The complex number z is given by $z = \frac{4+3\mathrm{i}}{2-\mathrm{i}}$. Express z in the Cartesian form $x + \mathrm{i}y$, where x and y are real numbers.

To express z in the form $x + \mathrm{i}y$, multiply the numerator and the denominator by the complex conjugate of the denominator. The denominator is $2 - \mathrm{i}$, so its complex conjugate is $2 + \mathrm{i}$. Multiply numerator and denominator: $z = \frac{(4+3\mathrm{i})(2+\mathrm{i})}{(2-\mathrm{i})(2+\mathrm{i})}$ Expand the numerator: $(4 + 3\mathrm{i})(2 + \mathrm{i}) = 8 + 4\mathrm{i} + 6\mathrm{i} + 3\mathrm{i}^2 = 8 + 10\mathrm{i} - 3 = 5 + 10\mathrm{i}$ Expand the denominator.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Modulus Equations & Linear Graphs

Solve the modulus equation $|3x - 5| = 2x + 1$ for all real solutions.

a. $x = 6$ **and** $x = 0.8$ ✓

b. $x = 6$ only

c. $x = 4$ and $x = -2$

d. $x = 0.8$ only

Case 1: $3x - 5 = 2x + 1 \Rightarrow x = 6$. Check: $|13| = 13$ (valid). Case 2: $3x - 5 = -(2x + 1) \Rightarrow 5x = 4 \Rightarrow x = 4/5 = 0.8$. Check: $|-2.6| = 2.6 = 2(0.8) + 1$ (valid). Both are solutions.

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22 [Foundational] • Target: Modulus Equations & Linear Graphs

Solve the modulus equation $|2x - 7| = |x + 2|$.

a. $x = 9$ **and** $x = 5/3$ ✓

b. $x = 9$ only

c. $x = 5$ and $x = -3$

d. $x = 3$ and $x = -1$

Square both sides: $(2x-7)^2 = (x+2)^2 \Rightarrow (2x-7-x-2)(2x-7+x+2) = 0 \Rightarrow (x-9)(3x-5) = 0 \Rightarrow x = 9$ or $x = 5/3$.

23 [Foundational] • Target: Modulus Equations & Linear Graphs

Find the coordinates of the vertex of the graph $y = |2x - 6| + 4$.

a. $(3, 4)$ ✓

b. $(-3, 4)$

c. $(3, -4)$

d. $(6, 4)$

The vertex occurs where the argument of the modulus equals zero: $2x - 6 = 0 \Rightarrow x = 3$. At $x = 3$, $y = 0 + 4 = 4$. Vertex is $(3, 4)$.

24 [Foundational] • Target: Modulus Inequalities

Solve the inequality $|2x - 5| < 7$.

a. $-1 < x < 6$ ✓

b. $x < -1$ or $x > 6$

c. $-6 < x < 1$

d. $1 < x < 6$

$-7 < 2x - 5 < 7 \Rightarrow -2 < 2x < 12 \Rightarrow -1 < x < 6$.

25 [Foundational] • Target: Modulus Inequalities

Solve the inequality $|3x + 1| \geq |x - 4|$.

a. $x \leq -5/2$ **or** $x \geq 3/4$ ✓

b. $-5/2 \leq x \leq 3/4$

c. $x \leq -1$ or $x \geq 4$

d. $x \geq 3/4$

Square both sides: $(3x+1)^2 \geq (x-4)^2 \Rightarrow (3x+1)^2 - (x-4)^2 \geq 0 \Rightarrow (4x-3)(2x+5) \geq 0$. Roots at $3/4$ and $-5/2$. Product is non-negative outside roots: $x \leq -5/2$ or $x \geq 3/4$.

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26 [Foundational] • Target: Modulus Inequalities

Find the set of values of x satisfying $|x-2| > 2x-1$.

a. $x < 1$ ✓

b. $x > 1$

c. $-1 < x < 1$

d. $x < 3$

For $x \geq 2$: $x-2 > 2x-1 \Rightarrow x < -1$ (no solution in $[2, \infty)$). For $x < 2$:
$-(x-2) > 2x-1 \Rightarrow -x+2 > 2x-1 \Rightarrow 3x < 3 \Rightarrow x < 1$.

27 [Foundational] • Target: Composite and Inverse Functions in A2

The function $f(x) = 2 + \ln(x-3)$ is defined for $x > 3$. Find the inverse function $f^{-1}(x)$ and state its domain.

a. $f^{-1}(x) = 3 + e^{x-2}$, **domain** $x \in \mathbb{R}$ ✓

b. $f^{-1}(x) = 3 - e^{x-2}$, domain $x > 0$

c. $f^{-1}(x) = 2 + e^{x-3}$, domain $x \in \mathbb{R}$

d. $f^{-1}(x) = \ln(x-2) + 3$, domain $x > 2$

Let $y = 2 + \ln(x-3) \Rightarrow \ln(x-3) = y-2 \Rightarrow x-3 = e^{y-2} \Rightarrow x = 3 + e^{y-2}$. The range of f is $\mathbb{R}$, so domain of f^{-1} is $\mathbb{R}$.

28 [Foundational] • Target: Composite and Inverse Functions in A2

Given $f(x) = e^{2x}$ for $x \in \mathbb{R}$ and $g(x) = \frac{1}{x}$ for $x > 0$, solve the equation $fg(x) = e^4$.

a. $x = 1/2$ ✓

b. $x = 2$

c. $x = 1/4$

d. $x = 4$

$fg(x) = f(1/x) = e^{2/x} = e^4 \Rightarrow 2/x = 4 \Rightarrow x = 2/4 = 1/2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Composite and Inverse Functions in A2

A function $h(x) = x^2 - 4x + 1$ is defined on the domain $x \le k$. Find the greatest value of k for which h is one-to-one.

a. $k = 2$ ✓

b. $k = 4$

c. $k = 0$

d. $k = 1$

The parabola has vertex at $x = -(-4)/(2) = 2$. For $x \le k$ to be strictly decreasing (one-to-one), the domain must not extend past the vertex, so greatest $k = 2$.

30 [Intermediate] • Target: Transformations & Analytical Modeling

Describe the transformation mapping the graph of $y = \ln(x)$ onto the graph of $y = \ln(3x - 6)$.

a. **Horizontal stretch scale factor** $1/3$ **followed by translation** $[2, 0]$ ✓

b. Translation $[6, 0]$ followed by horizontal stretch scale factor 3

c. Horizontal stretch scale factor 3 followed by translation $[-2, 0]$

d. Vertical stretch scale factor 3 followed by translation $[2, 0]$

Write as $y = \ln[3(x - 2)]$. The factor 3 represents a horizontal stretch by scale factor $1/3$, and $(x - 2)$ represents a translation of 2 units in the positive x-direction ($[2, 0]$).

31 [Intermediate] • Target: Transformations & Analytical Modeling

Find the number of real roots of the equation $|2x - 3| = x^2 - 2$.

a. **2 roots** ✓

b. 4 roots

c. 0 roots

d. 1 root

For $x \ge 1.5$: $2x - 3 = x^2 - 2 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$ (not in $x \ge 1.5$). For $x < 1.5$: $-2x + 3 = x^2 - 2 \Rightarrow x^2 + 2x - 5 = 0 \Rightarrow x = \frac{-2 \pm \sqrt{24}}{2} = -1 \pm \sqrt{6}$. $-1 + \sqrt{6} \approx 1.45 < 1.5$ (valid), and $-1 - \sqrt{6} \approx -3.45 < 1.5$ (valid). Exactly 2 roots.

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32 [Intermediate] • Target: Transformations & Analytical Modeling

The graph of $y = f(x)$ passes through $(2, 5)$. What are the coordinates of the corresponding point on $y = 3 - 2f(x + 1)$?

a. $(1, -7)$ ✓

b. $(3, -7)$

c. $(1, 13)$

d. $(3, 7)$

New $x = 2 - 1 = 1$. New $y = 3 - 2(5) = 3 - 10 = -7$. Point is $(1, -7)$.

33 [Intermediate] • Target: Modulus Equations & Linear Graphs

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ **or** $5 < k < 11$ ✓

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

1. **Find $f^{-1}(x)$ and its domain:** Completing the square gives $f(x) = (x - 3)^2 + 4$. For the domain $x \geq 3$, $f(x)$ is strictly increasing with range $[4, \infty)$. Setting $y = (x - 3)^2 + 4$ with $x \geq 3$ yields $x - 3 = \sqrt{y - 4}$, so $f^{-1}(x) = 3 + \sqrt{x - 4}, \quad \text{with domain } x \geq 4$. 2. **Set up the equation $f^{-1}(x) = |2x - k|$:** Let $u = \sqrt{x - 4} \geq 0$, which implies $x = u^2 + 4 \geq 4$. The equation becomes $3 + u = |2(u^2 + 4) - k| = |2u^2 + 8 - k|$. 3. **Analyz.

34 [Intermediate] • Target: Modulus Inequalities

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$ ✓

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

To find the inverse function $f^{-1}(x)$, set $y = f(x)$ and rearrange to express x in terms of y: $y = \frac{2x+1}{x-3}$ Multiply both sides by $x - 3$: $y(x - 3) = 2x + 1$ $yx - 3y = 2x + 1$ Collect all terms containing x on one side and constant/independent terms on the other: $yx - 2x = 3y + 1$ Factor out x: $x(y - 2) = 3y + 1$ Divide by $y - 2$: $x = \frac{3y+1}{y-2}$ Interchanging the variables gives: $f^{-1}(x) = \frac{3x+1}{x-2}$ Common Misconceptions: - Choice.

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35 [Intermediate] • Target: Composite and Inverse Functions in A2

Given $f(x) = x + 2$ and $g(x) = 2x + 3$, find $fg(-4)$.

a. -5

b. -2

c. **-3** ✓

d. -1

Applying Functions: Modulus & Inverse principles, $g(-4) = -5$, so $fg(-4) = f(-5) = -3$. Therefore, the verified correct option is -3.

36 [Intermediate] • Target: Transformations & Analytical Modeling

Given $f(x) = -3x - 11$, evaluate $f(|-6|)$.

a. **-29** ✓

b. -26

c. 29

d. 7

Applying Functions: Modulus & Inverse principles, $|-6| = 6$, so $f(|-6|) = f(6) = -29$. Therefore, the verified correct option is -29.

37 [Intermediate] • Target: Modulus Equations & Linear Graphs

Given $f(x) = -3x - 1$ and $g(x) = -4x + 13$, find $fg(0)$.

a. -1

b. 13

c. 17

d. **-40** ✓

Applying Functions: Modulus & Inverse principles, $g(0) = 13$, so $fg(0) = f(13) = -40$. Therefore, the verified correct option is -40.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 13.01 Continuous Random Variables

The function $f(x) = (x + 11)^2 - 4$ has domain $x \geq -11$. State the range of f.

a. $f(x) \leq -4$

b. $f(x) \geq -4$ ✓

c. $f(x) \geq -11$

d. $x \geq -11$

Since $x \geq -11$, the minimum of f occurs at $x = -11$, giving $f(-11) = -4$, and f increases thereafter. Range: $f(x) \geq -4$.

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39 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 14.01 Sampling & Estimation

Given $f(x) = 2x + 3$ and $g(x) = 3x + 3$, find $fg(4)$.

a. **33** ✓

b. 36

c. 11

d. 15

Applying Functions: Modulus & Inverse principles, $g(4) = 15$, so $fg(4) = f(15) = 33$. Therefore, the verified correct option is 33.

40 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 15.01 Hypothesis Testing

The function $f(x) = (x + 8)^2 - 10$ has domain $x \geq -8$. State the range of f.

a. $f(x) \geq -8$

b. $f(x) \geq -10$ ✓

c. $f(x) \leq -10$

d. $x \geq -8$

Since $x \geq -8$, the minimum of f occurs at $x = -8$, giving $f(-8) = -10$, and f increases thereafter. Range: $f(x) \geq -10$.

41 [Challenging] • Target: Modulus Equations & Linear Graphs • Cross-Lesson: 2.01 Exponentials & Logarithms

Given $f(x) = 3x + 8$, evaluate $f(|-5|)$.

a. **23** ✓

b. 26

c. -7

d. -23

Applying Functions: Modulus & Inverse principles, $|-5| = 5$, so $f(|-5|) = f(5) = 23$. Therefore, the verified correct option is 23.

42 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 3.01 Advanced Trigonometry

Given $f(x) = 5x + 8$, evaluate $f(|-9|)$.

a. **53** ✓

b. -53

c. -37

d. 56

Applying Functions: Modulus & Inverse principles, $|-9| = 9$, so $f(|-9|) = f(9) = 53$. Therefore, the verified correct option is 53.

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43 [Challenging] • Target: Composite and Inverse Functions in A2 • Cross-Lesson: 4.01 Differentiation

Given $f(x) = -4x - 11$, evaluate $f(|6|)$.

a. 35

b. **-35** ✓

c. -31

d. -32

Applying Functions: Modulus & Inverse principles, $|6| = 6$, so $f(|6|) = f(6) = -35$. Therefore, the verified correct option is -35.

44 [Challenging] • Target: Transformations & Analytical Modeling • Cross-Lesson: 5.01 Integration

Given $f(x) = -2x - 9$, evaluate $f(|4|)$.

a. **-17** ✓

b. 17

c. -14

d. -13

Applying Functions: Modulus & Inverse principles, $|4| = 4$, so $f(|4|) = f(4) = -17$. Therefore, the verified correct option is -17.

45 [Challenging] • Target: Modulus Equations & Linear Graphs • Cross-Lesson: 6.01 Sequences & Series

Given $f(x) = 3x + 1$ and $g(x) = -x - 5$, find $gf(-3)$.

a. -2

b. **3** ✓

c. -5

d. -8

Applying Functions: Modulus & Inverse principles, $f(-3) = -8$, so $gf(-3) = g(-8) = 3$. Therefore, the verified correct option is 3.

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